

CO2-AT2
ITA1407 - Ethical Hacking
Social Engineering Simulation

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Question 1

As a Security Analyst, perform an OSINT-based email enumeration assessment on the target domain `tesla.com` using theHarvester in Kali Linux. Collect publicly available email addresses, subdomains, hosts, and IP addresses associated with the organization. Categorize the discovered email addresses into likely roles such as Support, Sales, Human Resources, Administration, and Technical Teams. Analyze how the gathered information could assist an attacker during the reconnaissance phase and explain how it could be used for spear-phishing, credential harvesting, and impersonation attacks. Finally, recommend at least five countermeasures to reduce information leakage and social engineering risks.

Kali linux Commands :

`theHarvester -d tesla.com -b all`

`theHarvester -d tesla.com -b bing`

`theHarvester -d tesla.com -b crtsh`

`theHarvester -d tesla.com -b yahoo`

Command Explanation:

- **theHarvester -d tesla.com -b all** : Searches all supported public sources for email addresses, subdomains, hosts, and IP addresses related to the target domain.
- **theHarvester -d tesla.com -b bing** : Collects publicly available information using the Bing search engine.
- **theHarvester -d tesla.com -b crtsh** : Searches Certificate Transparency (crt.sh) logs to discover subdomains and certificate-related information.
- **theHarvester -d tesla.com -b yahoo** : Collects publicly available information using the Yahoo search engine.

Output Screenshot:

```
Session Actions Edit View Help
(joshitha@joshi)~$ theHarvester -d tesla.com -b rapiddns -l 100
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****
[*] Target: tesla.com
[*] Searching RapidDns.
[*] No IPs found.
[*] No emails found.
[*] No people found.
[*] Hosts found: 88
akamai-apigateway-automation-billing.tesla.com:akamai-apigateway-automation-billing.tesla.com.edgekey.net
akamai-apigateway-automation.tesla.com:akamai-apigateway-automation.tesla.com.edgekey.net
akamai-apigateway-captiveunderwriting.tesla.com:akamai-apigateway-captiveunderwriting.tesla.com.edgekey.net
akamai-apigateway-dev-warptmsapiserver.tesla.com:akamai-apigateway-dev-warptmsapiserver.tesla.com.edgekey.net
akamai-apigateway-logisticsratesapi.tesla.com:akamai-apigateway-logisticsratesapi.tesla.com.edgekey.net
akamai-apigateway-procuretopayapi.tesla.com:akamai-apigateway-procuretopayapi.tesla.com.edgekey.net
akamai-apigateway-shipmentplanningapi.tesla.com:akamai-apigateway-shipmentplanningapi.tesla.com.edgekey.net
akamai-apigateway-stg-captiveunderwriting.tesla.com:akamai-apigateway-stg-captiveunderwriting.tesla.com.edgekey.net
akamai-apigateway-stg-deliveryopsapi1.tesla.com:akamai-apigateway-stg-deliveryopsapi1.tesla.com.edgekey.net
akamai-apigateway-stg-finplateng-defi.tesla.com:akamai-apigateway-stg-finplateng-defi.tesla.com.edgekey.net
akamai-apigateway-stg-finplateng-routeone.tesla.com:akamai-apigateway-stg-finplateng-routeone.tesla.com.edgekey.net
akamai-apigateway-stg-fts.tesla.com:akamai-apigateway-stg-fts.tesla.com.edgekey.net
akamai-apigateway-stg-inventorytxnextapi.tesla.com:akamai-apigateway-stg-inventorytxnextapi.tesla.com.edgekey.net
akamai-apigateway-stg-translationpipeline.tesla.com:akamai-apigateway-stg-translationpipeline.tesla.com.edgekey.net
akamai-apigateway-vfx.tesla.com:akamai-apigateway-vfx.tesla.com.edgekey.net
akamai-apigateway-warptqapi.tesla.com:akamai-apigateway-warptqapi.tesla.com.edgekey.net
```

Question 2

As a cybersecurity awareness consultant, conduct a phishing awareness simulation using GoPhish and dummy employee accounts provided in the lab. Create an email template, launch a campaign, monitor user interactions, and generate a report showing delivery rate, email open rate, and click rate. Based on the results, provide recommendations to improve user awareness against phishing attacks.

KaliLinux Commands:

```
cd gophish
```

```
./gophish
```

Open the GoPhish dashboard in a web browser:

<https://127.0.0.1:3333>

Command Explanation:

- **cd gophish** : Changes the current directory to the GoPhish installation folder.
- **./gophish** : Starts the GoPhish phishing simulation server.
- **https://127.0.0.1:3333** : Opens the GoPhish web dashboard to create email templates, launch phishing campaigns, and view campaign reports.

Output:

GoPhish Server Started Successfully

Listening on:

https://127.0.0.1:3333

Admin Login Successful

Campaign Name : Phishing Awareness Campaign

Email Template : Password Expiry Notification

Target Group : Dummy Employees

Emails Sent : 50

Emails Delivered : 48

Emails Opened : 36

Links Clicked : 12

Campaign Status : Completed

Question 3

Using Nmap Aggressive Scan (-A) on scanme.nmap.org, gather detailed information about the target including operating system details, open ports, and service versions. Assume you are an attacker and describe how the collected information could be leveraged to conduct a social engineering attack against the IT support team. Prepare a report explaining the attack methodology and defensive recommendations.

KaliLinux Commands:

```
nmap -A scanme.nmap.org
```

(Optional)

```
nmap -A -Pn scanme.nmap.org
```

Command Explanation:

- **nmap -A scanme.nmap.org** : Performs an aggressive scan that detects the operating system, identifies open ports, detects service versions, runs default NSE scripts, and performs traceroute.

- **nmap -A -Pn scanme.nmap.org** : Performs the same aggressive scan while skipping host discovery by treating the target host as online.

Output Screenshot:

```

joshitha@joshi: ~
Session Actions Edit View Help
joshitha@joshi)~]
$ nmap scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 15:50 +0530
Stats: 0:00:13 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 84.53% done; ETC: 15:50 (0:00:02 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.28s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 992 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
25/tcp    filtered smtp
80/tcp    open  http
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 15.99 seconds

joshitha@joshi)~]
$ nmap -sV scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 15:51 +0530
Stats: 0:00:06 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 40.00% done; ETC: 15:52 (0:00:11 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.26s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 992 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  tcpwrapped
25/tcp    filtered smtp
80/tcp    open  http      Apache httpd 2.4.7 ((Ubuntu))
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open  nping-echo Nping echo
31337/tcp open  tcpwrapped

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 27.14 seconds

joshitha@joshi)~]
$ nmap -Oscanme.nmap.org
Unknown argument to -O.

```

```

joshitha@joshi: ~
Session Actions Edit View Help
joshitha@joshi)~]
$ nmap -sV scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 15:51 +0530
Stats: 0:00:06 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 40.00% done; ETC: 15:52 (0:00:11 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.26s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 992 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  tcpwrapped
25/tcp    filtered smtp
80/tcp    open  http      Apache httpd 2.4.7 ((Ubuntu))
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open  nping-echo Nping echo
31337/tcp open  tcpwrapped

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 27.14 seconds

joshitha@joshi)~]
$ nmap -Oscanme.nmap.org
Unknown argument to -O.

```

```

Session Actions Edit View Help

(joshitha@joshi)-[~]
$ nmap -A scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-04 15:54 +0530
Stats: 0:00:02 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 17.15% done; ETC: 15:54 (0:00:10 remaining)
Stats: 0:00:28 elapsed; 0 hosts completed (1 up), 1 undergoing Traceroute
Traceroute Timing: About 32.26% done; ETC: 15:54 (0:00:00 remaining)
Stats: 0:00:29 elapsed; 0 hosts completed (1 up), 1 undergoing Traceroute
Traceroute Timing: About 64.52% done; ETC: 15:54 (0:00:01 remaining)
Stats: 0:00:41 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 96.88% done; ETC: 15:54 (0:00:00 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.13s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb
2f
Not shown: 992 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linu
x; protocol 2.0)
|_ ssh-hostkey:
|_ 1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|_ 2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|_ 256 96:02:bb:9e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_ 256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
25/tcp    filtered smtp
80/tcp    open  http      Apache httpd 2.4.7 ((Ubuntu))
|_ http-title: Go ahead and ScanMe!
|_ http-favicon: Nmap Project
|_ http-server-header: Apache/2.4.7 (Ubuntu)
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open  nping-echo Nping echo
31337/tcp open  tcpwrapped
OS fingerprint not ideal because: Host distance (23 network hops) is greater th
an five
No OS matches for host
Network Distance: 5 hops
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 80/tcp)
HOP RTT      ADDRESS
1   24.81 ms  10.72.123.21
2   ...
3   76.23 ms  255.0.0.2
4   76.57 ms  255.0.0.3
5   40.91 ms  scanme.nmap.org (45.33.32.156)

```

```

3   76.23 ms  255.0.0.2
4   76.57 ms  255.0.0.3
5   40.91 ms  scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at http
s://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 42.61 seconds

(joshitha@joshi)-[~]
$

```